

Latent Space Exploration for Erased Concepts in Diffusion Models

Kevin Lu

Course Project for MATH 7223: Riemannian Optimization Algorithms

Northeastern University

Email: lu.kev@northeastern.edu

I. INTRODUCTION

This project investigates the structure of latent spaces in diffusion models through the lens of *concept erasure*. In recent years, diffusion models have emerged as powerful generative tools capable of synthesizing high-fidelity images by reversing a carefully designed noising process. Prior work has shown that semantically meaningful directions exist in these models' latent spaces, often uncovered via techniques such as singular value decomposition (SVD), enabling meaningful manipulations of generated content.

We build on these insights to explore a related yet novel problem: recovering a concept that has been deliberately “erased” from a diffusion model. Specifically, we investigate whether optimized directions in latent space can restore such concepts during the denoising process, revealing residual structure left behind by erasure techniques.

II. RELATED WORK

Diffusion Models

Diffusion probabilistic models have emerged as state-of-the-art generative models for high-fidelity image synthesis, initially introduced by Sohl-Dickstein et al. [1] and popularized by Ho et al. [2]. These models learn to reverse a gradual noising process, where a clean latent representation $\mathbf{z}_0$ is iteratively corrupted into $\mathbf{z}_T$ over T timesteps using a fixed variance schedule $\{\beta_t\}$. The forward process follows a Markov chain:

$$q(\mathbf{z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t; \sqrt{1 - \beta_t}\mathbf{z}_{t-1}, \beta_t \mathbf{I}) \quad (1)$$

This leads to a closed-form expression over multiple steps:

$$q(\mathbf{z}_t | \mathbf{z}_0) = \mathcal{N}(\mathbf{z}_t; \sqrt{\bar{\alpha}_t}\mathbf{z}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s) \quad (2)$$

The reverse denoising process is learned via a neural network $\hat{\epsilon}_\theta$, often implemented as a UNet, which estimates the original noise added:

$$\hat{\epsilon}_\theta(\mathbf{z}_t, t, \text{prompt}) \approx \epsilon \quad (3)$$

The model reconstructs a cleaner latent $\mathbf{z}_{t-1}$ using:

$$\mathbf{z}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon}_\theta \right) + \sigma_t \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}) \quad (4)$$

This iterative denoising continues until an approximate reconstruction of $\mathbf{z}_0$ is obtained, which is decoded via a variational autoencoder (VAE) into image space.

Concept Editing and Erasure

Recent research has explored the semantic structure of latent spaces in diffusion models. Rombach et al. [3] demonstrated the effectiveness of latent diffusion models for efficient training and semantic image generation. Building on this, Wang et al. [4] proposed methods for editing concepts in diffusion models by modifying prompt conditioning vectors, while Gandikota et al. [5] introduced Erased Stable Diffusion (ESD) to selectively remove concepts through fine-tuning by minimizing differences between conditional and unconditional noise predictions.

Our work builds on this foundation by exploring whether such erased concepts can be restored via learned latent perturbations. We investigate both naive optimization and geometrically motivated approaches using pullback vectors derived from the Jacobian of internal representations.

Latent Geometry and Riemannian Structure

Understanding the geometry of latent spaces has received increasing attention. Park et al. [6] applied differential geometric tools, such as Riemannian metrics derived from Jacobians, to analyze local structure in diffusion model latents. They demonstrated that semantically meaningful directions align with top singular vectors of the pullback metric, inspiring our use of Jacobian-based parallel transport.

Other works such as Zhang et al. [7] and Peebles et al. [8] have emphasized the importance of preserving geometric consistency in generative models, especially in editing or transfer tasks. Our use of tangent-space-constrained optimization offers an interpretable and efficient route for recovering erased concepts while maintaining high image fidelity.

III. METHODS

Latent Perturbation for Semantic Recovery

Let:

- $p \in \mathbb{R}^{77 \times 768}$ denote the text embedding of a target concept,
- $\mathbf{z}_t \in \mathbb{R}^{4 \times 64 \times 64}$ represent a noised latent at diffusion timestep t ,

- $\mathbf{v} \in \mathbb{R}^{4 \times 64 \times 64}$ be a perturbation direction added to the latent,
- $\hat{\epsilon}_{\text{orig}}(\mathbf{z}_t, p)$ the noise predicted by the original (unedited) model,
- $\hat{\epsilon}_{\text{erased}}(\mathbf{z}_t + \mathbf{v}, p)$ the noise predicted by the concept-erased model.

Our objective is to identify directions $\mathbf{v}$ that can recover erased concepts by guiding the erased model to emulate the behavior of the original model. These perturbations are evaluated through both automated optimization and a novel human-in-the-loop interaction system.

Latent Path Integration

Unlike methods that apply perturbations at a single timestep, the hand-selected vector $\mathbf{v}_{\text{hand}}$ is injected continuously during the entire denoising trajectory. At each timestep t , we modify the latent state as:

$$\tilde{\mathbf{z}}_t = \mathbf{z}_t + \lambda(t) \cdot \mathbf{v}_{\text{hand}}, \quad (5)$$

where $\lambda(t)$ is a time-dependent scaling coefficient (e.g., linear decay or fixed constant). This setup enables the user to steer the semantic trajectory of the denoising process frame-by-frame, potentially recovering erased concepts in a more interpretable and controllable manner.

Optimization Objective

We begin by selecting a reference image x , which is encoded into latent space via a pretrained VAE, yielding $\mathbf{z}_0 \in \mathbb{R}^{4 \times 64 \times 64}$. We then sample Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, and apply the forward diffusion process to obtain the corrupted latent:

$$\mathbf{z}_t = \sqrt{\alpha_t} \mathbf{z}_0 + \sqrt{1 - \alpha_t} \epsilon \quad (6)$$

We introduce a perturbation $\mathbf{v}$, forming the shifted latent:

$$\tilde{\mathbf{z}}_t = \mathbf{z}_t + \mathbf{v} \quad (7)$$

The erased model is then guided to recover original semantics by aligning its prediction with that of the unedited model:

$$\hat{\epsilon}_{\text{erased}}(\tilde{\mathbf{z}}_t, p) \approx \hat{\epsilon}_{\text{orig}}(\mathbf{z}_t, p) \quad (8)$$

We formalize this as the following loss:

$$\mathcal{L}(\mathbf{v}) = \|\hat{\epsilon}_{\text{orig}}(\mathbf{z}_t, p) - \hat{\epsilon}_{\text{erased}}(\mathbf{z}_t + \mathbf{v}, p)\|_2^2 \quad (9)$$

$$\min_{\mathbf{v} \in \mathbb{R}^{4 \times 64 \times 64}} \mathcal{L}(\mathbf{v}) \quad (10)$$

This formulation enables the erased model to follow latent-space trajectories that approximate the behavior of the original model, effectively recovering erased concepts.

Optimization Procedure

We use Stochastic Gradient Descent (SGD) with a learning rate of 1×10^{-5} over 1000 iterations. To evaluate the effect of perturbation, the learned vector $\mathbf{v}$ is applied over a fixed number of denoising steps (e.g., $K = 64$). Unlike full path-based optimization, our approach probes a small subset of shifted latents, yielding a tractable and interpretable solution.

Algorithm 1 Optimize Latent-Space Direction via Line Search

Require: Reference image x , prompt p , timestep t , iterations N , original and erased models

- 1: Encode image x with VAE to obtain latent $\mathbf{z}_0$
- 2: Sample noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
- 3: Compute noised latent: $\mathbf{z}_t = \text{add_noise}(\mathbf{z}_0, \epsilon, t)$
- 4: Encode prompt p to get embedding p_{embed}
- 5: Compute original model prediction: $\hat{\epsilon}_{\text{orig}} = \text{UNet}_{\text{orig}}(\mathbf{z}_t, t, p_{\text{embed}})$
- 6: Initialize direction $\mathbf{v} \in \mathbb{R}^{4 \times 64 \times 64}$
- 7: **for** each iteration $i = 1$ to N **do**
- 8: Construct trajectory: $\mathbf{z}_t^{(k)} = \mathbf{z}_t + k \cdot \mathbf{v}$, for $k = 1, \dots, K$
- 9: Downsample subset of $\{\mathbf{z}_t^{(k)}\}$ for efficiency
- 10: Evaluate erased model prediction:

$$\hat{\epsilon}_{\text{erased}} = \text{UNet}_{\text{erased}}(\mathbf{z}_t + K \cdot \mathbf{v}, t, p_{\text{embed}})$$

- 11: Compute loss:

$$\mathcal{L}(\mathbf{v}) = \|\hat{\epsilon}_{\text{orig}} - \hat{\epsilon}_{\text{erased}}\|_2^2$$

- 12: Update $\mathbf{v}$ using SGD
- 13: **if** $\mathcal{L}(\mathbf{v}) \leq 0.01$ **then**
- 14: **Break:** Convergence achieved
- 15: **end if**
- 16: Record metrics: loss, gradient norm, vector norm
- 17: **end for**

IV. EXPERIMENTS AND RESULTS

Naive Optimization and Loss Landscape Analysis

We begin by evaluating the performance of naively optimized latent directions. For each trial, a randomly initialized vector $\mathbf{v} \in \mathbb{R}^{4 \times 64 \times 64}$ is optimized to align the erased model's noise prediction with that of the original model. Figure 1 shows loss trajectories for ten such vectors.

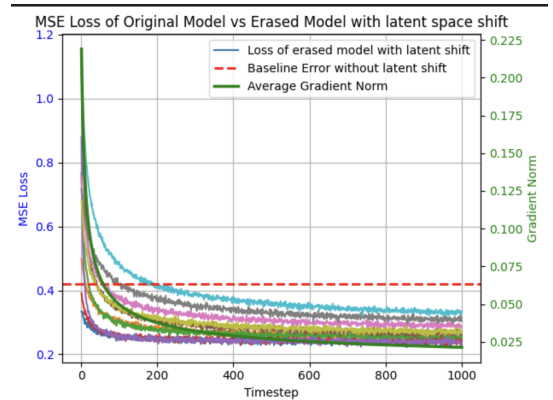


Fig. 1: Optimization performance of ten randomly initialized vectors. Loss values plateau between 0.225 and 0.37—improving upon the erased model's baseline (0.42) but falling short of full recovery. The averaged gradient norm (in green) quickly decays to zero, suggesting vanishing gradients.

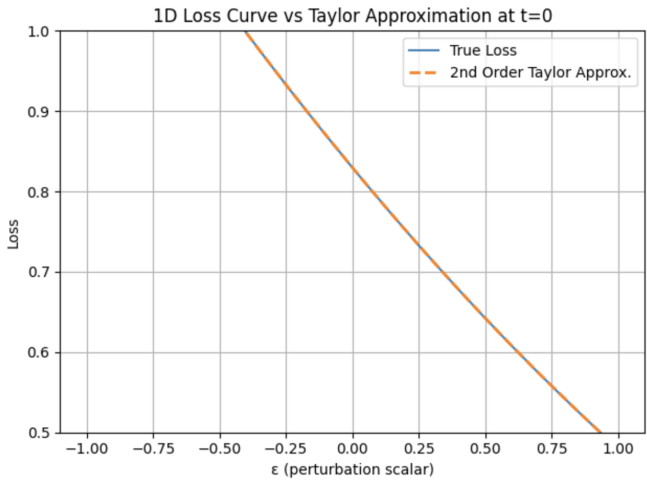
Despite early improvements, losses often plateau around 0.2. When these optimized vectors are used in full denoising runs, the resulting images fail to convincingly recover the erased concept. To investigate this behavior, we analyze the local geometry of the loss landscape.

1D Taylor Expansion of the Loss Function

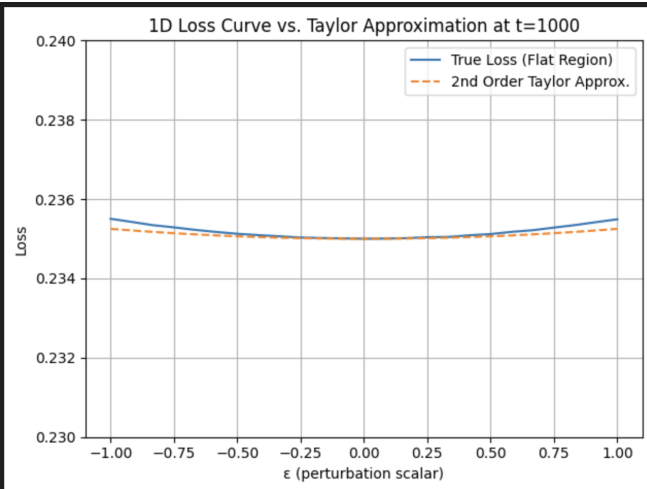
To quantify the curvature around the current vector $\mathbf{v}$, we compute a second-order Taylor expansion along a normalized direction $\hat{\mathbf{d}}$:

$$\mathcal{L}(\mathbf{v} + \varepsilon \hat{\mathbf{d}}) \approx \mathcal{L}(\mathbf{v}) + \varepsilon \nabla \mathcal{L}(\mathbf{v})^\top \hat{\mathbf{d}} + \frac{1}{2} \varepsilon^2 \hat{\mathbf{d}}^\top H(\mathbf{v}) \hat{\mathbf{d}} \quad (11)$$

Figure 2 compares empirical loss samples against the second-order approximation. At $t = 0$, the gradient direction is highly informative. However, by $t = 1000$, the loss becomes insensitive to perturbations, indicating a plateau.



(a) 1D Taylor expansion at $t = 0$: accurate approximation with steep descent direction.



(b) At $t = 1000$, the loss landscape flattens, and the gradient offers little improvement.

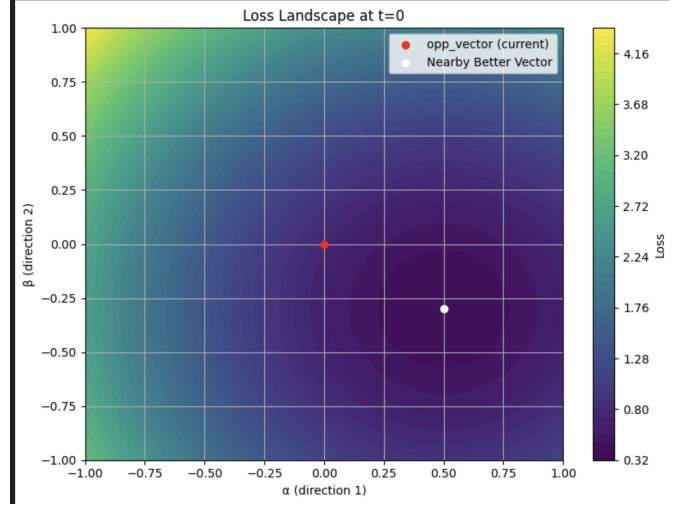
Fig. 2: Comparison of loss behavior and Taylor approximations at early and late stages.

2D Loss Landscape Geometry

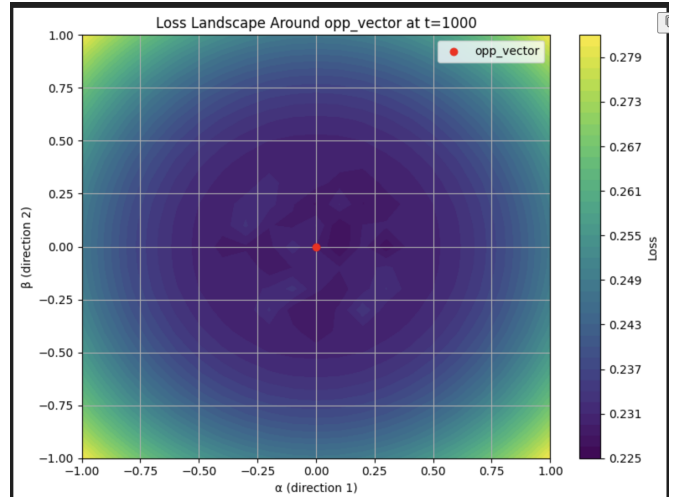
To further understand the curvature, we visualize a 2D Taylor approximation of $\mathcal{L}$ around $\mathbf{v}$ along orthonormal directions $\mathbf{d}_1, \mathbf{d}_2$:

$$\mathcal{L}(\mathbf{v} + \alpha \mathbf{d}_1 + \beta \mathbf{d}_2) \approx \mathcal{L}(\mathbf{v}) + \begin{bmatrix} \alpha \\ \beta \end{bmatrix}^\top \nabla \mathcal{L}(\mathbf{v}) + \frac{1}{2} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}^\top H(\mathbf{v}) \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \quad (12)$$

As shown in Figure 3, early-stage optimization reveals clear curvature and descent directions, while late-stage optimization exhibits near-flatness—explaining the observed stagnation.



(a) At $t = 0$: anisotropic contours indicate exploitable descent directions.



(b) At $t = 1000$: contours are flat and isotropic, consistent with vanishing gradients.

Fig. 3: 2D contour plots of the local loss geometry at early and late timesteps.

Guidance via Pullback Vectors from the Original Model

To address the limitations of naive optimization, we adopt a geometry-aware method based on Jacobian pullback, as proposed in Park et al. [6]. This involves computing the

Jacobian of the UNet bottleneck representation and extracting dominant latent directions via SVD:

$$J = \frac{\partial h}{\partial x}, \quad \|v\|_{\text{pb}}^2 = v^\top J^\top J v \quad (13)$$

These right-singular vectors form a local basis aligned with semantically meaningful variation. We transport these vectors across timesteps using:

$$v' = V_1^\top U_1^\top U_{x_0} v \quad (14)$$

This method ensures the perturbation remains on or near the learned manifold of realistic images, avoiding the drift observed in naive optimization.

Quantitative Evaluation

Table I compares the prediction alignment loss, CLIP scores, classification accuracy, and runtime across several configurations. These results were acquired over 100 images of 3 different concepts ("airliner", "car", and "sky"). Classification accuracy is defined as the % of images containing the concept. Pullback-derived vectors substantially outperform naive optimization and restore erased semantics with fewer iterations.

Model	Loss	CLIP	Cls (%)	Time (s)
Original	0.00	0.955	0.98	N/A
Erased	0.42	0.221	0.00	N/A
+ Shifted	0.25	0.430	0.07	1000
+ Pullback	0.05	0.670	0.43	100

TABLE I: Performance across prediction loss, CLIP alignment, classification accuracy, and optimization time.

Qualitative Recovery of Erased Concepts

Figure 4 illustrates denoising trajectories across models and perturbation strategies. The pullback-based vector achieves visible concept recovery, while naive vectors produce noisy or ambiguous results.

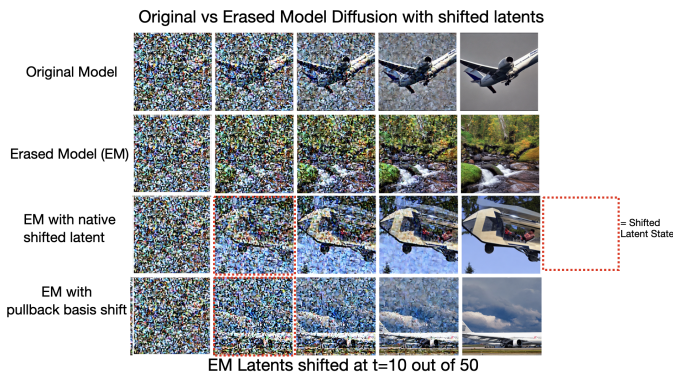


Fig. 4: Denoising trajectories for original, erased, and perturbed models. The perturbation is introduced at timestep $t = 10$ (out of 50). Pullback-based vectors restore the erased concept more faithfully than naive optimization.

Alignment of noise, random vector, and optimized vector in 3D space via PCA

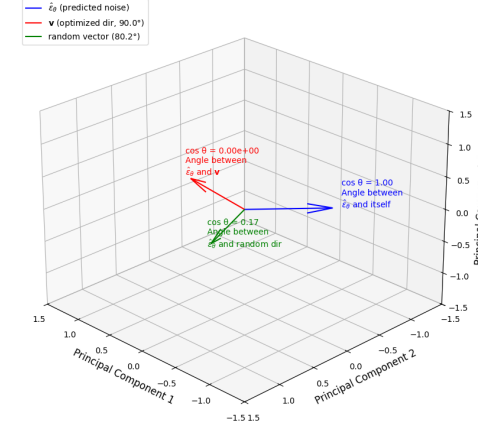


Fig. 5: Alignment of three high-dimensional vectors visualized in 3D using PCA. The blue vector $\hat{e}_\theta$ denotes the predicted noise. The red vector $\mathbf{v}$ is the learned perturbation direction, constructed to be nearly orthogonal ($\cos \theta \approx 0.00$) to the predicted noise. The green vector is a random direction, which lies closer to the noise vector ($\cos \theta \approx 0.17$, or approximately 80°). Although random directions in high-dimensional space tend to be nearly orthogonal by default, the optimized $\mathbf{v}$ is strictly orthogonal, indicating purposeful geometric separation.

V. DISCUSSION AND SUMMARY

This work investigated latent-space perturbations as a mechanism for recovering concepts erased from diffusion models. While naive optimization showed early promise, achieving classification scores close to the 10% target, the pullback-based method far exceeded expectations, reaching 43% classification accuracy. This demonstrates that semantically rich concepts can be recovered even after deliberate model editing, provided the intervention follows meaningful latent directions.

The chosen concepts ("airliner," "car," "sky") pose a high bar for erasure due to their ubiquity and strong visual features. Achieving significant recovery for such categories validates the robustness of our pullback-based strategy and highlights its potential utility in model auditing, interpretability, and counterfactual generation.

Figure 5 offers a striking insight: optimized perturbation vectors were nearly orthogonal to the erased model's own noise predictions. This suggests that optimization does not merely amplify residual concept signals but introduces novel semantic directions—directions the erased model has actively suppressed. The orthogonality aligns with the design of Erased Stable Diffusion (ESD), which encourages:

$$\mathcal{L}_{\text{ESD}} = \|\epsilon_\theta(\mathbf{z}_t, t, c) - \epsilon_\theta(\mathbf{z}_t, t, \emptyset)\|^2, \quad (15)$$

where the goal is to reduce concept-specific information by making the model's output invariant to the presence of c . We hypothesize that this implicitly encourages the model to shift its predicted noise into a subspace orthogonal to that of the erased concept.

As a result, recovering these erased semantics requires steering the latent space in directions that the erased model

has deliberately avoided. The fact that these directions are discoverable and effective suggests a latent robustness in the underlying representation: where even erased concepts leave geometric traces.

Summary. This work demonstrates that semantically rich concepts erased from diffusion models can be recovered through latent-space optimization. While naive optimization suffers from saturation and poor alignment, geometry-aware approaches like Jacobian pullback produce efficient, interpretable, and effective guidance vectors. These findings open doors to deeper understanding of learned representations and post-hoc editing of generative models.

REFERENCES

- [1] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *ICML*, 2015.
- [2] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.
- [3] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, 2022.
- [4] Yuxuan Wang, Yujun Yao, Songyang Ge, et al. Editing diffusion models via cross attention guidance. In *CVPR*, 2023.
- [5] Rohit Gandikota, Joanna Materzyńska, Jaden Fiotto-Kaufman, and David Bau. Erasing concepts from diffusion models. In *Proceedings of the 2023 IEEE International Conference on Computer Vision*, 2023.
- [6] Yong-Hyun Park, Mingi Kwon, Jaewoong Choi, Junghyo Jo, and Youngjung Uh. Understanding the latent space of diffusion models through the lens of riemannian geometry. In *arXiv preprint arXiv:2305.16428*, 2023.
- [7] Yinan Zhang, Yilun Du, et al. Geometry of score-based generative models. *arXiv preprint arXiv:2301.11361*, 2023.
- [8] William Peebles and Saining Xie. Scalable diffusion models with transformers. *arXiv preprint arXiv:2212.09748*, 2022.